

DBMS – PAYMENT MANAGEMENT

Vilvah

1. Database Design

1.1 Payment Table

Attribute	Data Type	Constraint
Payment_ID	NUMBER	Primary Key
Order_ID	NUMBER	Foreign Key
Transaction_ID	VARCHAR2(50)	UNIQUE
Payment_Method	VARCHAR2(30)	NOT NULL
Payment_Date	DATE	NOT NULL
Amount	NUMBER(10,2)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL

The Payment table stores payment information associated with each order, including payment mode, date, amount, and transaction status.

2. Table Creation

2.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER(10) PRIMARY KEY,  
3     Order_ID NUMBER(10) NOT NULL,  
4     Payment_Method VARCHAR2(20),  
5     Transaction_ID VARCHAR2(50) UNIQUE,  
6     Payment_Status VARCHAR2(20),  
7     Payment_Date DATE,  
8     Amount NUMBER(10,2) NOT NULL,  
9     CONSTRAINT FK_PAYMENT_ORDER  
10    FOREIGN KEY (Order_ID)  
11    REFERENCES Orders(Order_ID)  
12 );  
  
Table created.
```

3. Payment Data Insertion

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, A
mount)
  3 VALUES
  4 (1, 101, 'UPI', 'TXN10001', 'Success', DATE '2026-09-20', 798.00);

1 row created.

SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, A
mount)
  3 VALUES
  4 (2, 102, 'Card', 'TXN10002', 'Success', DATE '2026-09-21', 599.00);

1 row created.

SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, A
mount)
  3 VALUES
  4 (3, 103, 'UPI', 'TXN10003', 'Pending', DATE '2026-09-22', 449.00);

1 row created.

SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, A
mount)
  3 VALUES
  4 (4, 104, 'COD', 'TXN10004', 'Success', DATE '2026-09-23', 899.00);

1 row created.

SQL>
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID, Payment_Status, Payment_Date, A
mount)
  3 VALUES
  4 (5, 105, 'Card', 'TXN10005', 'Failed', DATE '2026-09-24', 699.00);

1 row created.
```

4. Manage Payment Transactions

4.1 Display Successful Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Success';

PAYMENT_ID  ORDER_ID  PAYMENT_METHOD
-----
TRANSACTION_ID                                PAYMENT_STATUS
-----
PAYMENT_D    AMOUNT
-----
          1      101 UPI
TXN10001
20-SEP-26      798
          2      102 Card
TXN10002
21-SEP-26      599
          4      104 COD
TXN10004
23-SEP-26      899
```

4.2 Display Failed Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Failed';

PAYMENT_ID ORDER_ID PAYMENT_METHOD
-----
TRANSACTION_ID PAYMENT_STATUS
-----
PAYMENT_ID AMOUNT
-----
          5      105 Card
TXN10005
24-SEP-26      699 Failed
```

5. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Success'
  3 WHERE Payment_ID = 1;

1 row updated.

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 1;

PAYMENT_ID ORDER_ID PAYMENT_METHOD
-----
TRANSACTION_ID PAYMENT_STATUS
-----
PAYMENT_ID AMOUNT
-----
          1      181 UPI
TXN10001
20-SEP-26      798 Success

SQL> UPDATE Payment
  2 SET Payment_Status = 'Failed'
  3 WHERE Payment_ID = 5;

1 row updated.

SQL> SELECT
  2 Payment_ID,
  3 Order_ID,
  4 Payment_Method,
  5 Transaction_ID,
  6 Payment_Status,
  7 Payment_Date,
  8 Amount
  9 FROM Payment
 10 WHERE Payment_Status IN ('Success', 'Failed')
 11 ORDER BY Payment_ID;

PAYMENT_ID ORDER_ID PAYMENT_METHOD
-----
TRANSACTION_ID PAYMENT_STATUS
-----
PAYMENT_ID AMOUNT
-----
          1      181 UPI
TXN10001
20-SEP-26      798 Success
          2      182 Card
TXN10002
21-SEP-26      599 Success
          3      183 UPI
TXN10003
22-SEP-26      449 Success
          4      184 CDD
TXN10004
23-SEP-26      899 Success
          5      185 Card
TXN10005
24-SEP-26      699 Failed

SQL> SELECT
  2 Payment_ID,
  3 Order_ID,
  4 Payment_Method,
  5 Transaction_ID,
  6 Payment_Status,
  7 Payment_Date,
  8 Amount
  9 FROM Payment
 10 WHERE Payment_Status IN ('Success', 'Failed')
 11 ORDER BY Payment_ID;

PAYMENT_ID ORDER_ID PAYMENT_METHOD
-----
TRANSACTION_ID PAYMENT_STATUS
-----
PAYMENT_ID AMOUNT
-----
          1      181 UPI
TXN10001
20-SEP-26      798 Success
          2      182 Card
TXN10002
21-SEP-26      599 Success
          3      183 UPI
TXN10003
22-SEP-26      449 Success
          4      184 CDD
TXN10004
23-SEP-26      899 Success
          5      185 Card
TXN10005
24-SEP-26      699 Failed
```

6. Count transactions by payment method

```
SQL> SELECT
  2   Payment_Method,
  3   COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Method
  6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
COD	1
Card	2
UPI	2

6.1 Calculate total amount by payment method

```
SQL> SELECT
  2   Payment_Method,
  3   SUM(Amount) AS Total_Amount
  4 FROM Payment
  5 GROUP BY Payment_Method
  6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
COD	899
Card	1298
UPI	1247

6.2 Analyze only successful payments

```
SQL> SELECT
  2   Payment_Method,
  3   COUNT(*) AS Successful_Transactions,
  4   SUM(Amount) AS Total_Amount
  5 FROM Payment
  6 WHERE Payment_Status = 'Success'
  7 GROUP BY Payment_Method
  8 ORDER BY Payment_Method;
```

PAYMENT_METHOD	SUCCESSFUL_TRANSACTIONS	TOTAL_AMOUNT
COD	1	899
Card	1	599
UPI	2	1247

7. Payment method with customer details

```
SQL> SELECT
  2   o.Customer_ID,
  3   p.Payment_Method,
  4   p.Payment_Status,
  5   p.Amount
  6 FROM Payment p
  7 JOIN Orders o
  8   ON p.Order_ID = o.Order_ID
  9 ORDER BY o.Customer_ID;
```

CUSTOMER_ID	PAYMENT_METHOD	PAYMENT_STATUS	AMOUNT
1	UPI	Success	798
2	Card	Success	599
3	UPI	Success	449
4	COD	Success	899
5	Card	Failed	699

7.1 Payment Transaction Report

```
SQL> SELECT
  2   Payment_ID,
  3   Order_ID,
  4   Transaction_ID,
  5   Payment_Method,
  6   Payment_Date,
  7   Payment_Status,
  8   Amount
  9 FROM Payment
 10 ORDER BY Payment_Date;
```

PAYMENT_ID	ORDER_ID	TRANSACTION_ID	PAYMENT_METHOD	PAYMENT_D	PAYMENT_STATUS	AMOUNT
1	101	TXN10001	UPI	20-SEP-26	Success	798
2	102	TXN10002	Card	21-SEP-26	Success	599
3	103	TXN10003	UPI	22-SEP-26	Success	449
4	104	TXN10004	COD	23-SEP-26	Success	899
5	105	TXN10005	Card	24-SEP-26	Failed	699

7.2 Payment Report with Customer ID

```
SQL> SELECT
  2   p.Payment_ID,
  3   o.Customer_ID,
  4   p.Order_ID,
  5   p.Transaction_ID,
  6   p.Payment_Method,
  7   p.Payment_Date,
  8   p.Payment_Status,
  9   p.Amount
 10 FROM Payment p
 11 JOIN Orders o
 12   ON p.Order_ID = o.Order_ID
 13 ORDER BY p.Payment_Date;
```

PAYMENT_ID	CUSTOMER_ID	ORDER_ID	TRANSACTION_ID	PAYMENT_METHOD
1	1	101	TXN10001	UPI
20-SEP-26	Success			
2	2	102	TXN10002	Card
21-SEP-26	Success			
3	3	103	TXN10003	UPI
22-SEP-26	Success			
4	4	104	TXN10004	COD
23-SEP-26	Success			
5	5	105	TXN10005	Card
24-SEP-26	Failed			

7.3 Payment Status Summary Report

```
SQL> SELECT
2   Payment_Status,
3   COUNT(*) AS Total_Transactions,
4   SUM(Amount) AS Total_Amount
5 FROM Payment
6 GROUP BY Payment_Status
7 ORDER BY Payment_Status;

PAYMENT_STATUS    TOTAL_TRANSACTIONS TOTAL_AMOUNT
-----
Failed              1             699
Success             4            2745

SQL> COMMIT;

Commit complete.
```